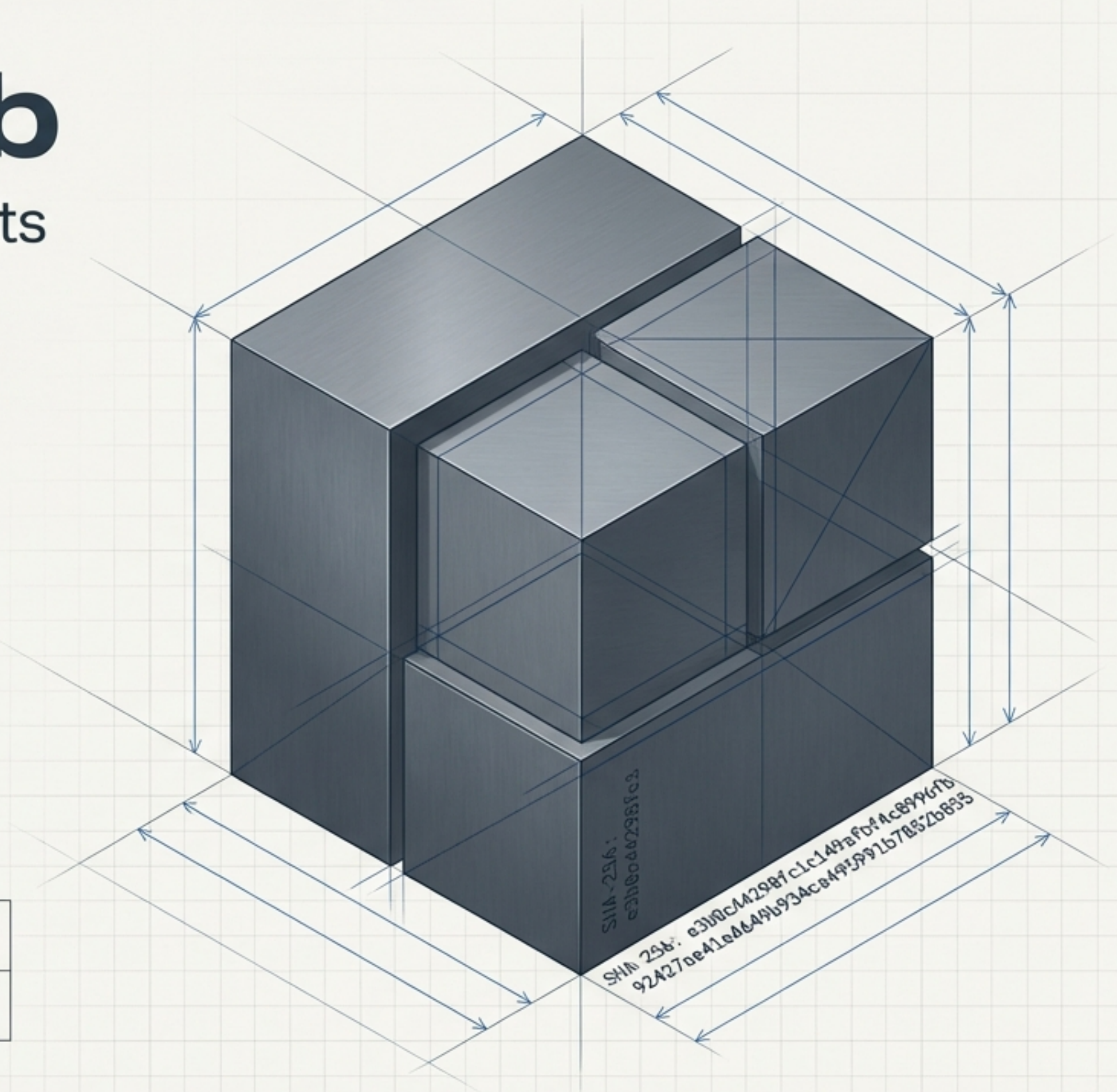


SG Vault Hub

The GitHub for Encrypted Vaults



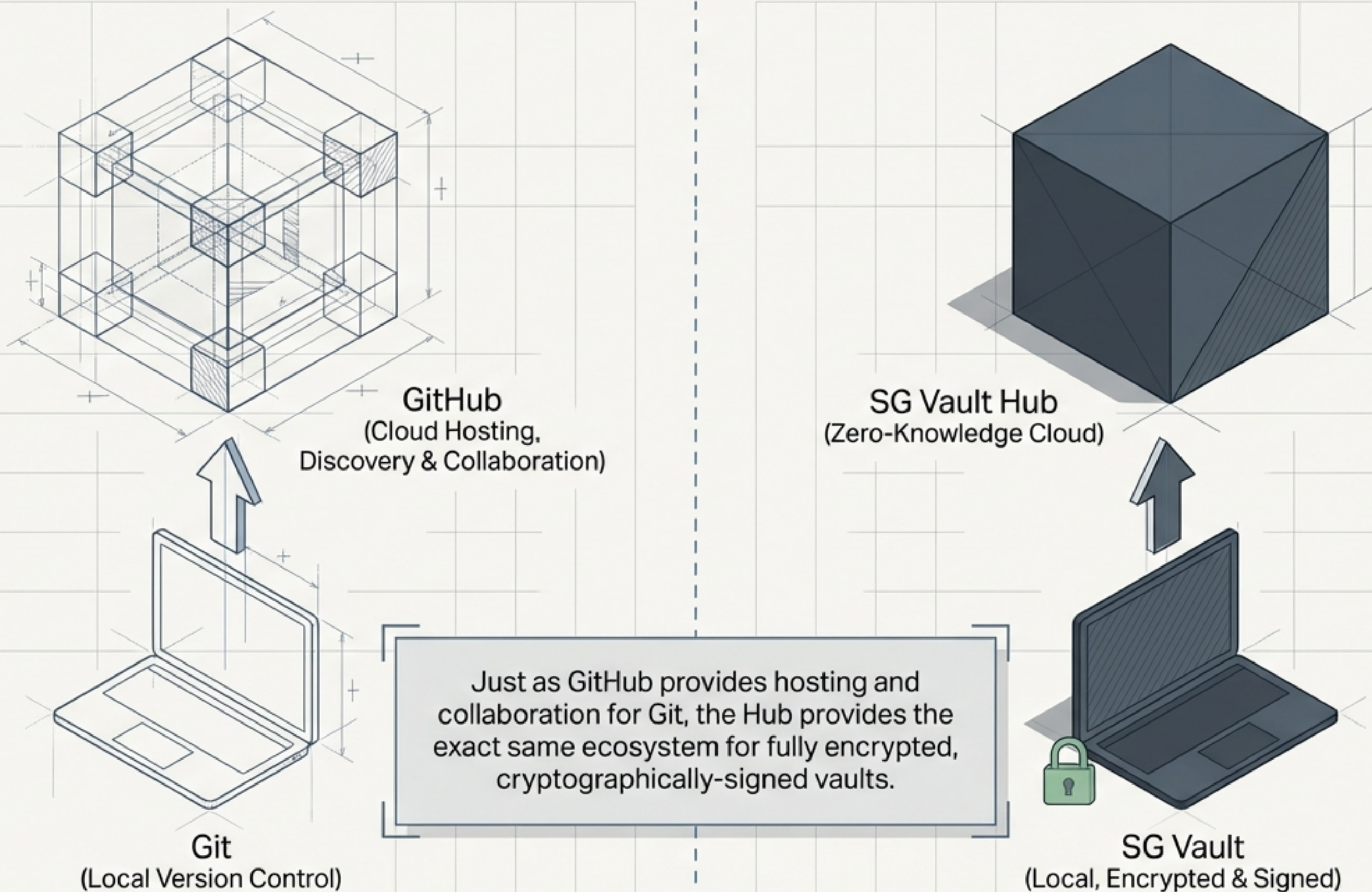
v0.13.30

Status: Vision Document

14 March 2026

Audience: Internal Stakeholders

THE CORE ANALOGY: ELEVATING THE TRUST MODEL

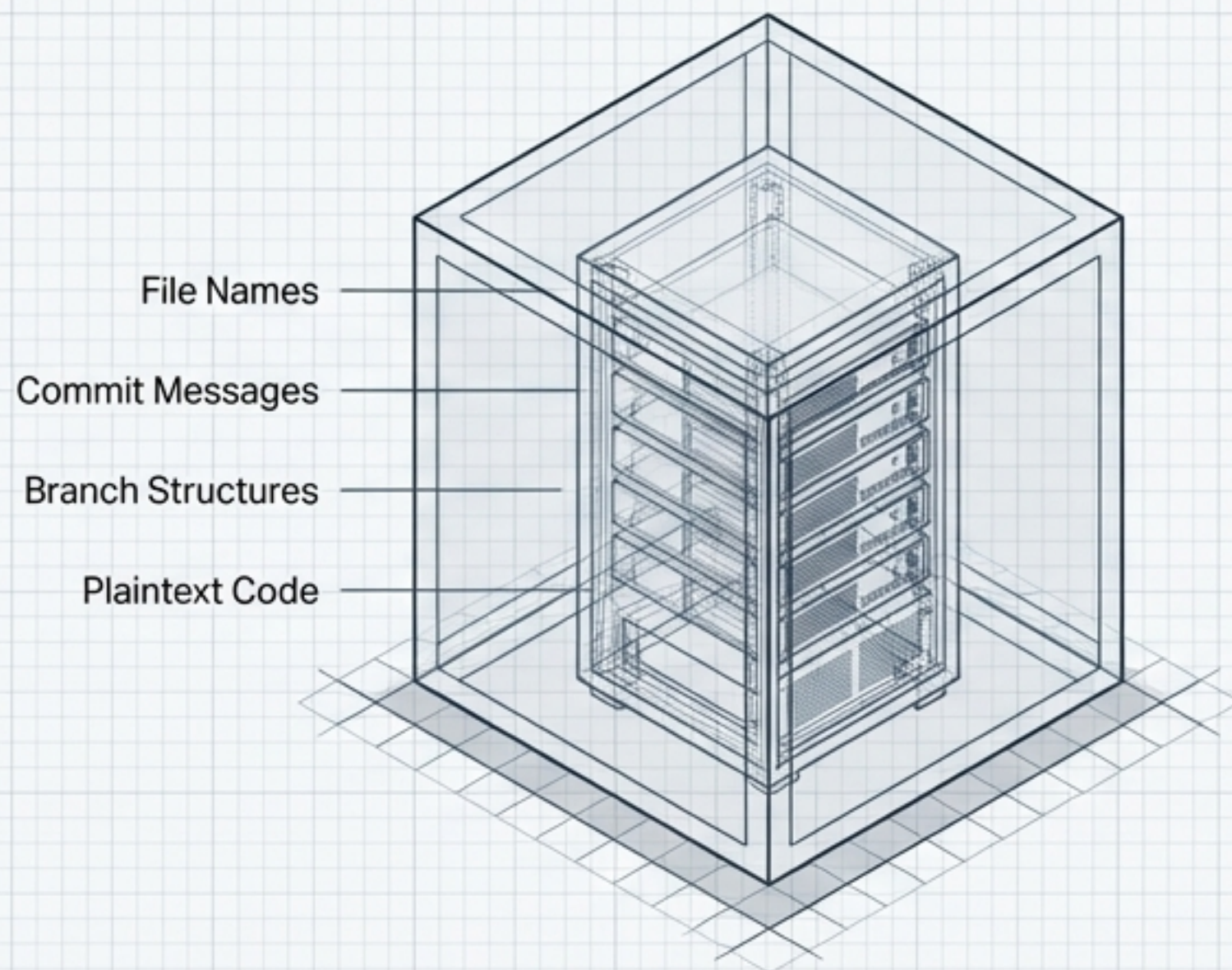


ZERO-KNOWLEDGE HOSTING

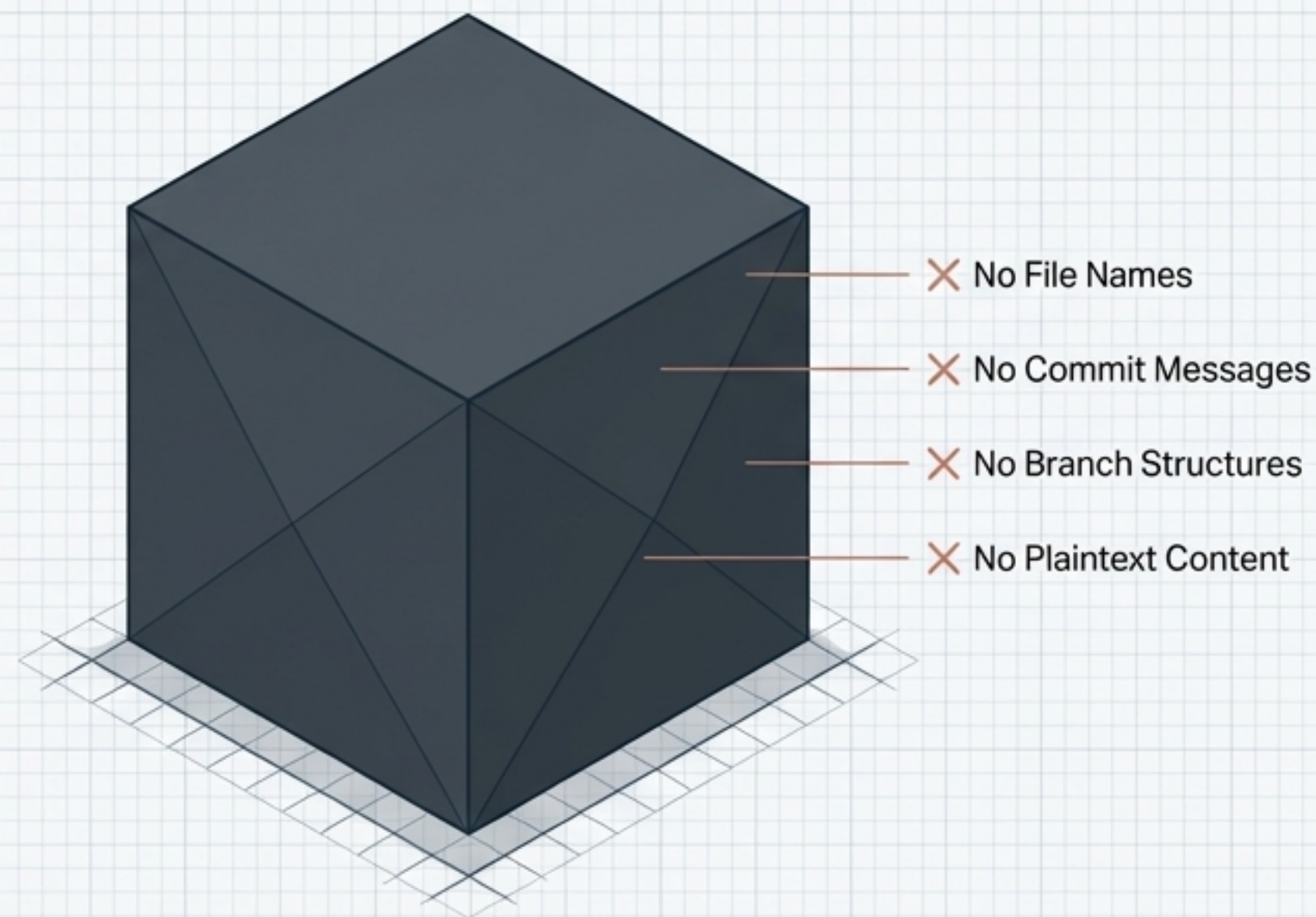
KEY INSIGHT PANEL

The Hub stores only encrypted blobs. The server cannot read vault contents. It facilitates key exchange, but never holds decrypted keys.

THE GLASS BOX



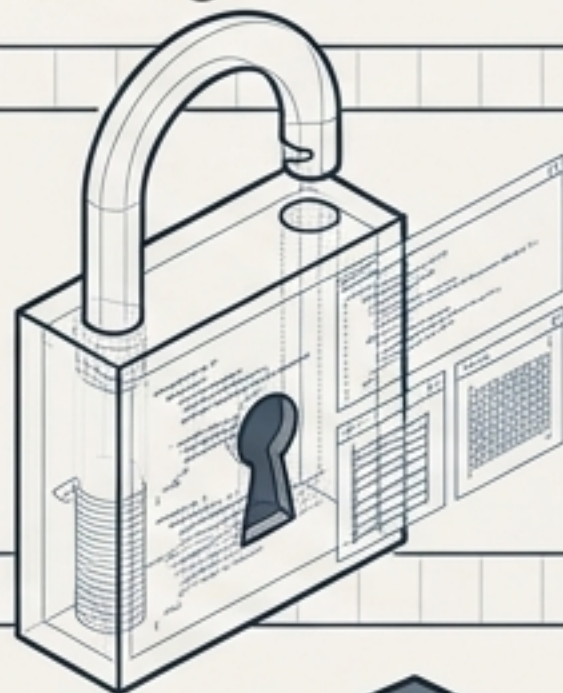
THE BLACK BOX



THE MAGIC OF CHANGE PACKS

Stronger than a Pull Request. Contributors submit **self-contained, signed blobs** to a secure pending slot without ever cloning the vault or accessing its keys. The vault becomes a **write-only** append target.

THE PULL REQUEST MODEL



Forks entire repository. Contributor has full read access to the codebase.

THE CHANGE PACK MODEL



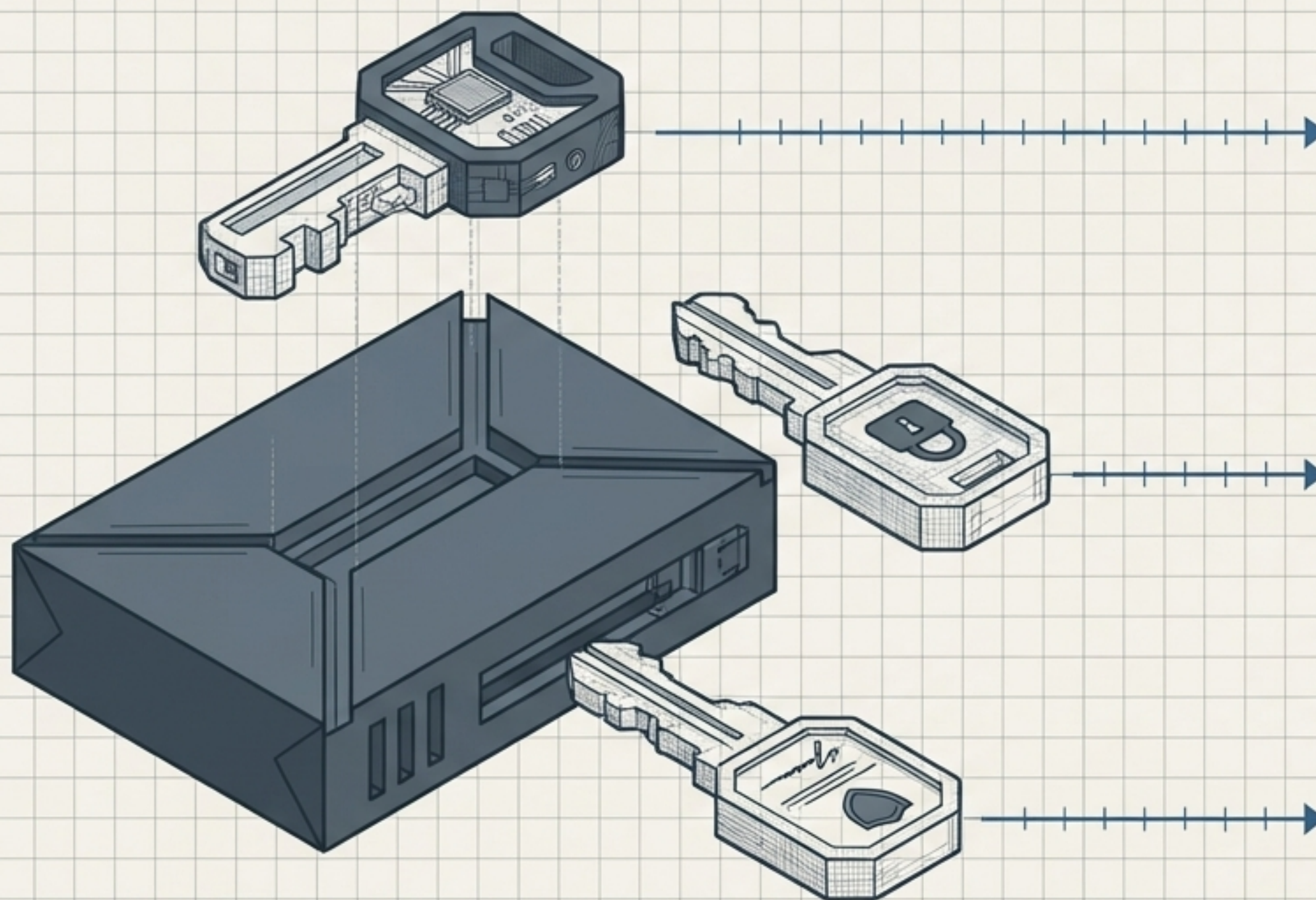
Packages encrypted change.
Zero knowledge of vault
contents required.

bare/pending/



ANATOMY OF A KEYLESS CONTRIBUTION

Zero cloning. Zero reading.
Absolute verification.



1. Scoped Write-Only Token

Grants permission for a single PUT request directly to the bare/pending/ directory.

PERMISSION: PUT /BARE/PENDING/ ONLY
TYPE: ONE-TIME USE
REVOCABLE: YES

2. Branch Public Key

Used by the contributor purely to encrypt their submission, ensuring only the vault owner can read it.

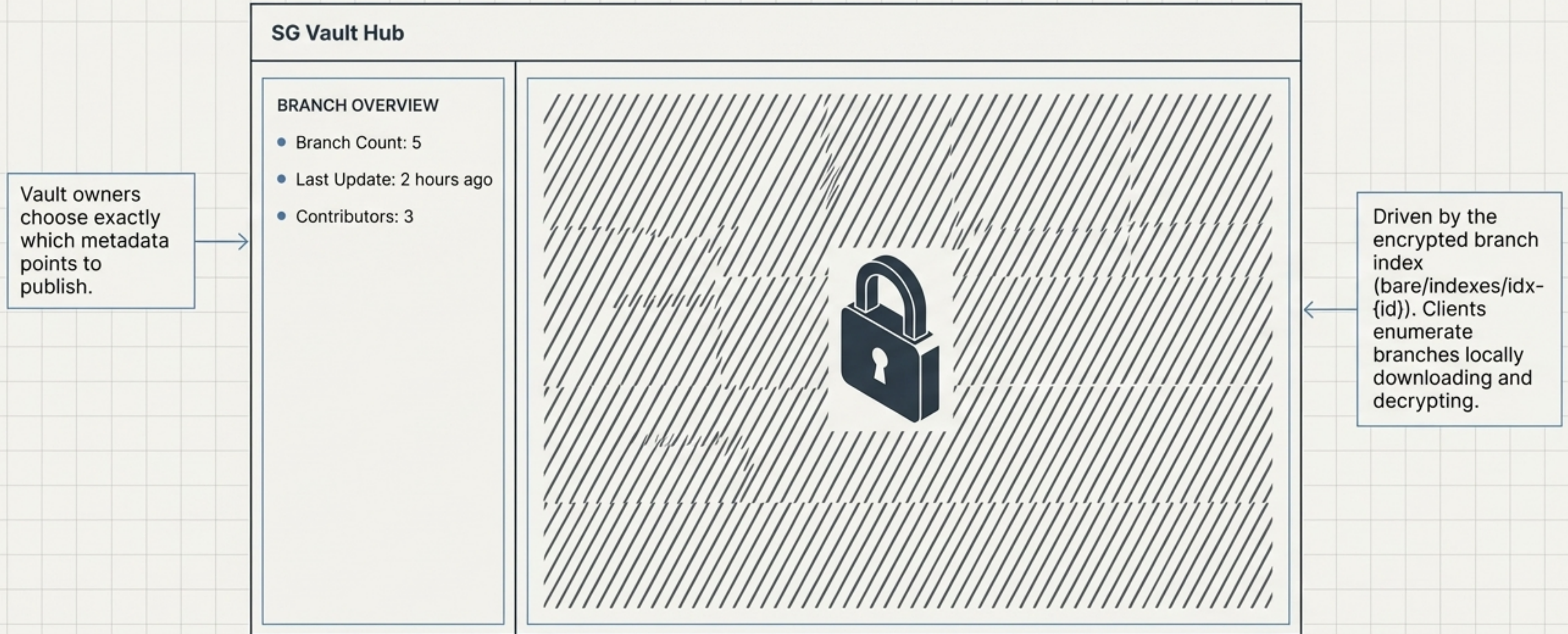
PURPOSE: ASYMMETRIC ENCRYPTION
PRIVATE KEY: SECURELY HELD BY VAULT OWNER
VISIBILITY: PUBLICLY AVAILABLE

3. Contributor Signing Key

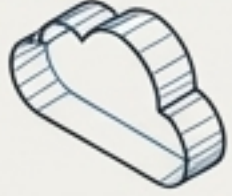
The cryptographic proof of provenance, verifying exactly who submitted the change pack.

VERIFICATION: DIGITAL SIGNATURE ALGORITHM (DSA)
PROVENANCE: HASH-BASED IDENTIFIER
STATUS: ABSOLUTE VERIFICATION

BRANCH VISIBILITY: METADATA WITHOUT EXPOSURE



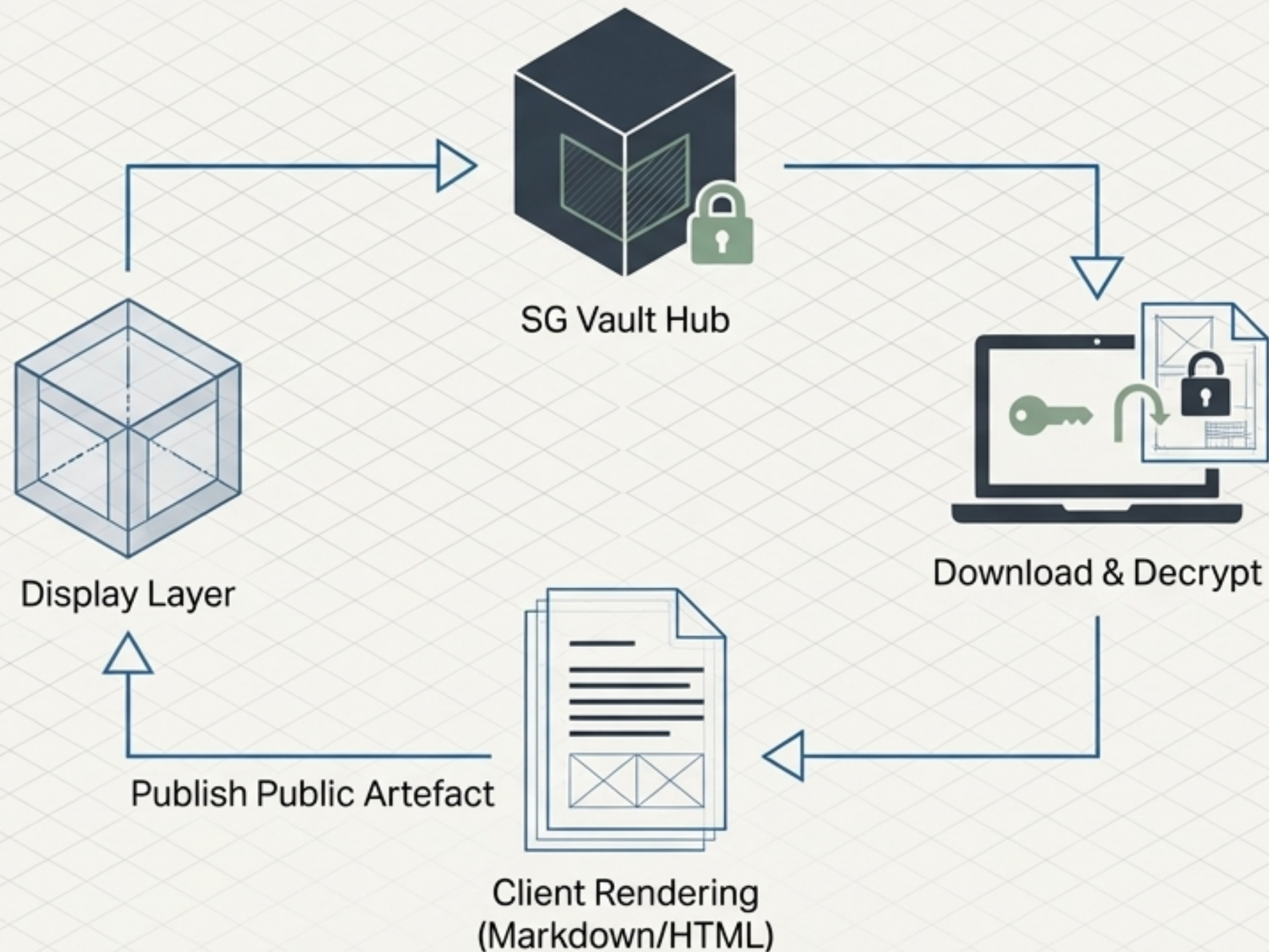
THE TRUST TRADE-OFF MATRIX

	 GitHub	 SG Vault Hub
Server reads repo contents	Yes	No
Server knows file names	Yes	No
Server knows branch names	Yes	No
Server knows commit messages	Yes	No
Search across repo content	Full-text	Not possible (encrypted)
Contributor identity	Account-based	Cryptographic (branch keys)
History integrity	Platform-guaranteed	Self-verifiable

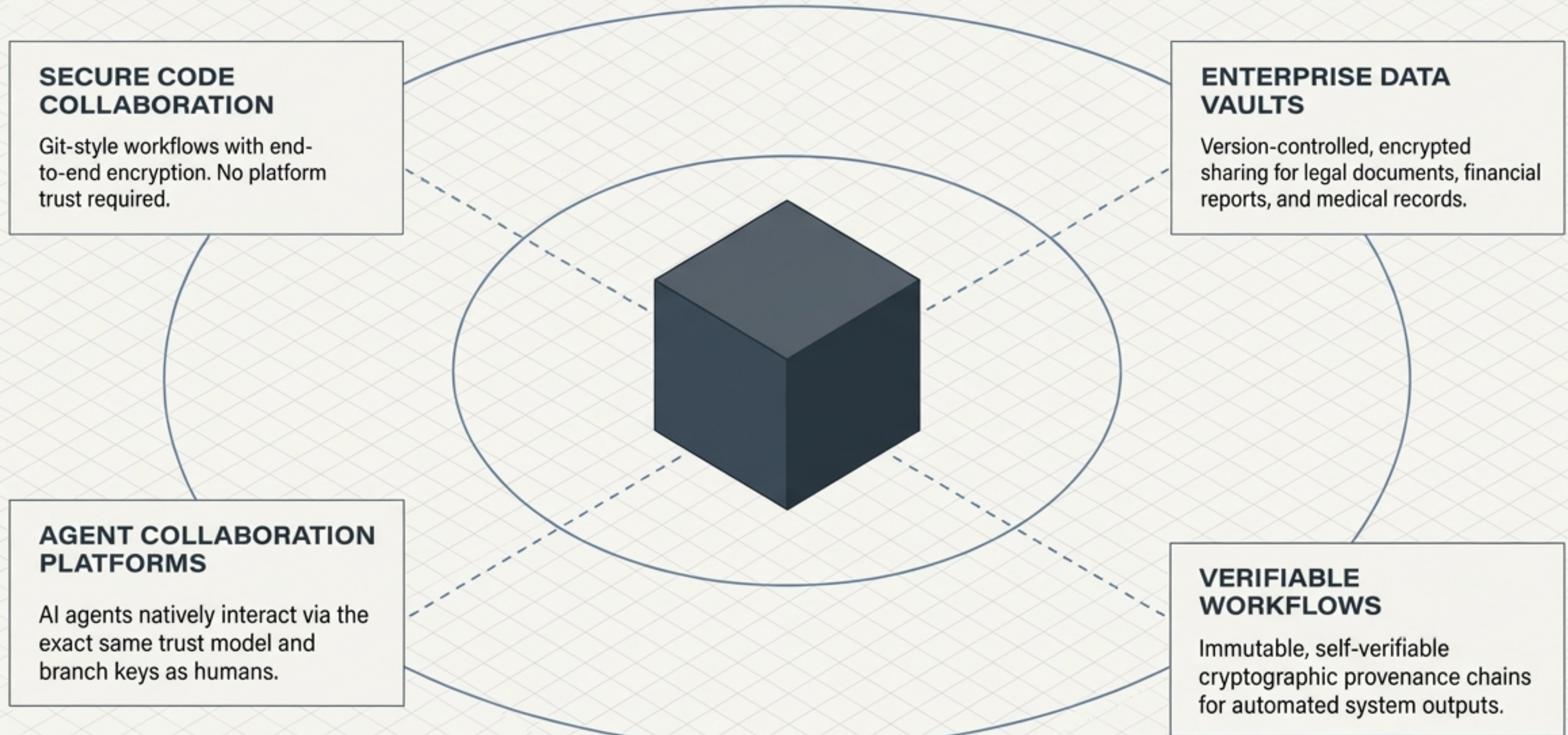
The trade-off is explicit: we sacrifice server-side content awareness to ensure trust resides entirely in the cryptography, never in the platform.

OPT-IN PUBLIC VIEWS: DECOUPLING CONTENT FROM HOSTING

Rendered READMEs or file listings are never generated automatically. The owner decrypts locally, renders the views, and explicitly pushes them to the Hub as separate public artefacts.

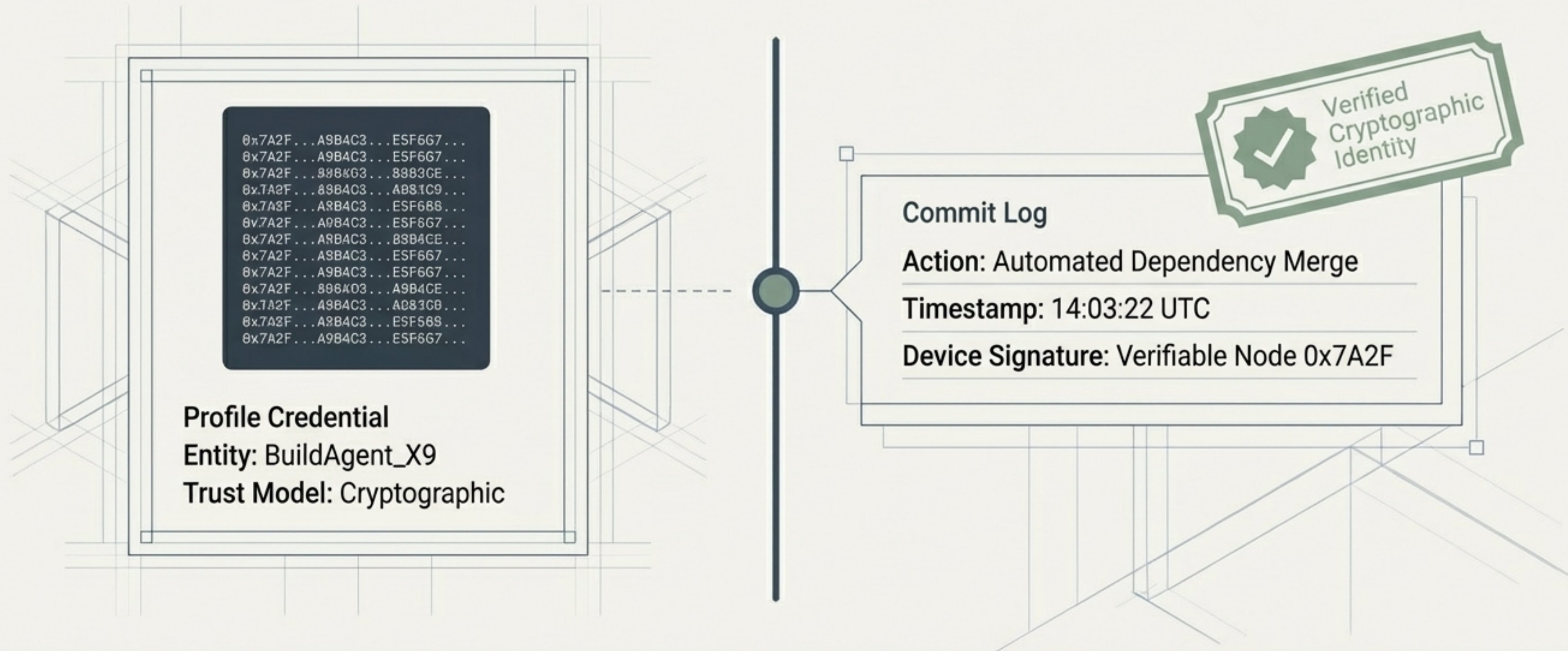


ECOSYSTEM ORBIT: BEYOND SOURCE CODE

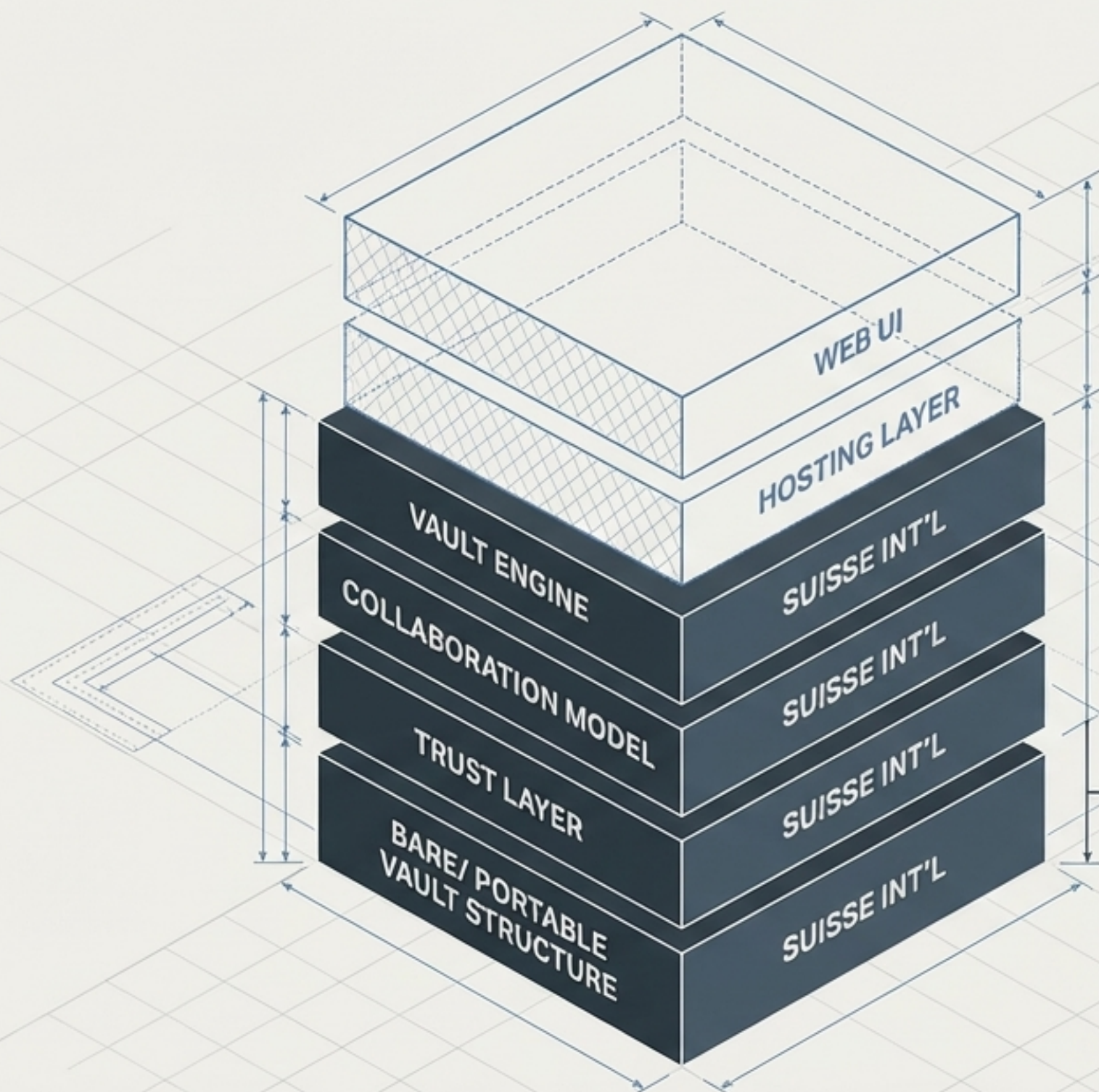


FIRST-CLASS IDENTITY FOR AI AGENTS

An AI agent contributing to a vault operates on the exact same trust model as a human engineer. The commit signature proves which agent, on which device, at what time—embedding an unbreakable provenance chain directly into the vault history.



WHERE WE ARE & NEXT STEPS



**BUILDING THE FUTURE
OF VERIFIABLE,
ZERO-KNOWLEDGE
COLLABORATION.**

SG Vault Hub is a natural extension, not a separate product from scratch. The backend sync logic already exists for any storage supporting read, write, delete, and list. Implementation requires only the hosting layer and the web interface.